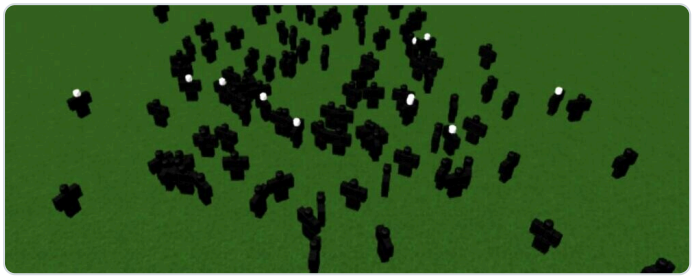
Gameplay Systems

Gameplay programmer, combat and AI scripter

`scenario-roblox-gameplay`

Builds NPC AI, melee and projectile combat and constraint cars for Roblox, with the server deciding every hit, rules tested in Lune and server cost measured in a playtest.



Live capture, Studio 0.743: 100 R15 NPCs on one scheduler, server 60 fps, AI 0.14 ms per frame.



Constraint car with spring suspension, source of the kit's car numbers. Suphi Kaner, Cars With Suspension, thumbnail.

USE IT WHEN

- Enemies that patrol, chase and attack, or 100 NPCs that lag the server
- Melee combos, guns or projectiles where hits register twice or pass through walls
- Cars with suspension that flip or steer wrong; replacing BodyVelocity with modern movers

WHAT THE EXPERTS INSIST ON

- Decide melee hits on the server with a box query, never Touched, and count one hit per character, never per limb. Ludius (scripting teacher)
- Spherecast, Blockcast and Shapecast ignore parts they start inside: check the start with an overlap, then sweep before moving. veraktt (scripter)
- A path counts only when ComputeAsync's pcall succeeds AND Status is Success; otherwise handle the failure. BrawlDev (Roblox developer since 2016)
- Stun cancels pending hits: drop the victim's queued swing and refuse its attack requests for a bounded stun. ArTheDev (combat scripter)
- Per wheel, one CylindricalConstraint and one SpringConstraint on the same attachments, upper limit below 0; keep the center of mass low. Suphi Kaner (Roblox scripter and teacher)

WORKFLOW, WITH THE GATE BEFORE EACH NEXT STAGE

1. **Install and probe.** **Gate:** Hook and remotes live in the playtest; Spherecast from inside returns the far part.
2. **Brain in Lune.** **Gate:** State machine and NPC brain specs pass in Lune before any playtest.
3. **Crowd and cost.** **Gate:** 0 of 100 NPC roots client-owned; server fps and p95 recorded at each count.
4. **Hunt.** **Gate:** The NPC reaches the player behind the wall and lands a hit.
5. **Combat.** **Gate:** Front dummies hit, back one untouched; spammed swings land at least 0.24 s apart.
6. **Vehicle.** **Gate:** Driver owns the chassis; W for 3 s covers 30+ studs; stays upright.
7. **Audit and review.** **Gate:** GameplayAudit reports 0 errors; captures read on one contact sheet.

KEY NUMBERS

100 R15 NPCs, measured in Studio	**server 60 fps, p95 17.6 ms, AI 0.14 ms per frame**
Scheduler defaults	**10 thinks per NPC per second, 30 paths per second**
Humanoid MoveTo timeout	**8 s, so re-issue MoveTo on every think**
Combo damage and minimum gaps	**10, 12, 20 damage; 0.30, 0.30, 0.45 s**
Knockback held 0.2 s, measured	**LinearVelocity moved a dummy 10.5 studs, one ApplyImpulse 1.2**
Car suspension (Suphi Kaner)	**Stiffness 3500, Damping 200, FreeLength 2.8, upper limit -1**

CHECKS THE AGENT RUNS ON ITS OWN WORK

- Lune specs pass; rbx_check and GameplayAudit report 0 errors.
- NpcService.stats(): no client-owned NPC roots, AI ms per frame recorded at target count.
- Dummy health and hit counts prove one hit per character per swing.
- Car samples for distance, heading and UpVector.Y; paths and query boxes drawn in captures.

HANDOFFS

Receives From architecture: the round loop, the binder for tagged weapons and hazards, and the damage service slot.

Delivers Hit rules to networking, NPC states to animation, hit points to VFX, input actions to UI, bench numbers to performance.

TOOLKIT `RBX_GAMEPLAY.PY`

- `install()`: writes kit and boot scripts into the open place
- `unit_tests()`: runs the Lune brain specs, returns pass counts
- `audit_place()`: offline GameplayAudit of a place file through Lune

MAIN EXPERT SOURCES

Suphi Kaner
Roblox scripter and teacher, Cars With Suspension

BrawlDev
Roblox developer since 2016, tutorial series, Pathfinding; Raycasting, guns, visualizers, shapecasts

Miller, Roblox Learn
software engineer at Roblox, Guard NPCs using a state machine

Ludius
scripting teacher, How To Script Hitboxes Properly

STATUS

Tests 20 Lune tests and 3 audit specs pass

In app 23 live tests pass in Roblox Studio 0.743

G8 gameplay systems

without skills	9/15
with skills	15/15